<u>AI Practical Work (MCA 2nd)</u>:

<u>Guideline:</u> For an AI practical lab manual using PROLOG/LISP, the "Requirements" section should clearly list what a student needs before performing the experiment. Typically, it includes:

Software setup:

- *SWI-PROLOG or any standard PROLOG interpreter installed.*
- *Text editor (VS Code, Sublime, Notepad++).*

Hardware setup:

- *A computer system with minimum 4 GB RAM and stable OS (Windows/Linux).*

Knowledge prerequisites:

- *Understanding of PROLOG syntax (facts, rules, queries).*
- *Basic logic programming concepts.*

Experiment materials:

- *The program file containing facts and rules (e.g., capital_of(nepal, kathmandu).).*
- *Queries to test the program.*

Documentation requirements:

- *Lab manual should include Objectives, Requirements, Problem Statement, Procedure, Code, Output, Observation, Conclusion.*

❖ ***Prepare a Lab Manual for the Following Using PROLOG/LIPS***

Simple question-answering

1. Write a PROLOG program to answer the question: Is Kathmandu the capital of Nepal?
2. Create a LISP program that answers: Is 7 greater than 5?

Family relation problems

3. Write a PROLOG program to represent family relations and query: Who is the father of Sita?
4. Develop a LISP program to check if two people are siblings.

Numerical problems

5. Write a PROLOG program to generate prime numbers up to 50.
6. Create a LISP program to compute the GCD of two numbers.

Logic-gate programs

7. Write a PROLOG program to simulate the AND gate.
8. Develop a LISP program to implement the truth table of OR gate.

Fundamental search techniques

9. Write a PROLOG program to perform depth-first search on a simple graph.
10. Create a LISP program to perform breadth-first search on a tree.

Menu-driven programs

11. Write a PROLOG program with a menu to calculate area of circle, rectangle, or triangle.
12. Develop a LISP program with a menu to perform addition, subtraction, multiplication, and division.

Inference and reasoning problems

13. Write a PROLOG program to infer if a student is eligible for scholarship based on marks and income.
14. Create a LISP program to reason whether a person can vote based on age and citizenship.

Tower of Hanoi

15. Write a PROLOG program to solve the Tower of Hanoi problem with 3 disks.
16. Develop a LISP program to display the steps of Tower of Hanoi for 4 disks.